EXPERIMENT 3

AIM: To design a Flutter user interface by using common Flutter widgets such as Text, Container, Row, Column, Image, TextField, ElevatedButton, Icon, and Scaffold.

OBJECTIVE: To understand and implement commonly used Flutter widgets for creating an interactive and user-friendly mobile application interface.

COURSE OUTCOME: Design and implement user interfaces using Flutter widgets, layouts, and animations, ensuring responsiveness and interactivity.

THEORY: Flutter is an open-source UI framework developed by Google for creating cross-platform mobile applications using the Dart programming language. Flutter uses widgets as the basic building blocks of its user interface. Common widgets such as Scaffold, AppBar, Text, Container, Row, Column, Icon, TextField, and Buttons are used to create and arrange different elements on the screen. Row arranges widgets horizontally, while Column arranges them vertically. Container is used for styling and spacing, and TextField accepts user input.
Flutter also provides Hot Reload, which allows developers to view code changes quickly without restarting the entire application. Thus, common Flutter widgets can be combined to create a simple, interactive, and user-friendly UI.

CODE:

```
import 'package:flutter/material.dart';

void main() => runApp(const PremiumApp());

class PremiumApp extends StatefulWidget {
  const PremiumApp({super.key});

  @override
  State<PremiumApp> createState() =>
_PremiumAppState();
}

class _PremiumAppState extends
State<PremiumApp> {
  bool _isDark = false;

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Nova Wallet',
      debugShowCheckedModeBanner: false,
      themeMode: _isDark ? ThemeMode.dark :
ThemeMode.light,
      theme: ThemeData(
        colorScheme:
ColorScheme.fromSeed(seedColor: const
Color(0xFF6366F1), brightness: Brightness.light),
        scaffoldBackgroundColor: const
Color(0xFFF8FAFC),
        useMaterial3: true,
      ),
      darkTheme: ThemeData(
        colorScheme:
ColorScheme.fromSeed(seedColor: const
Color(0xFF6366F1), brightness: Brightness.dark),
        scaffoldBackgroundColor: const
Color(0xFF0B0F19),
        useMaterial3: true,
      ),
      home: MainDashboard(isDark: _isDark,
onToggleTheme: (v) => setState(() => _isDark =
v)),
    );
  }
}

class MainDashboard extends StatefulWidget {
  final bool isDark;
  final ValueChanged<bool> onToggleTheme;

  const MainDashboard({super.key, required
this.isDark, required this.onToggleTheme});

  @override
  State<MainDashboard> createState() =>
_MainDashboardState();
}

class _MainDashboardState extends
State<MainDashboard> {
  int _navIndex = 0;
  double _balance = 12480.50;
  String _selectedCat = 'All';
  final List<String> _cats = ['All', 'Crypto', 'Stocks',
```

```
'NFTs'];

  final List<Map<String, dynamic>> _assets = [
    {'name': 'Bitcoin', 'symbol': 'BTC', 'val':
'\$64,230.00', 'change': '+4.2%', 'up': true, 'icon':
Icons.currency_bitcoin, 'color': Colors.orange},
    {'name': 'Ethereum', 'symbol': 'ETH', 'val':
'\$3,450.20', 'change': '+2.8%', 'up': true, 'icon':
Icons.token, 'color': Colors.deepPurple},
    {'name': 'Solana', 'symbol': 'SOL', 'val':
'\$145.80', 'change': '-1.4%', 'up': false, 'icon':
Icons.flash_on, 'color': Colors.teal},
  ];

  void _toast(String msg) {
    ScaffoldMessenger.of(context).clearSnackBars()
;
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text(msg), behavior:
SnackBarBehavior.floating, duration: const
Duration(seconds: 2)),
    );
  }

  @override
  Widget build(BuildContext context) {
    final theme = Theme.of(context);
    final isDark = widget.isDark;

    return Scaffold(
      appBar: AppBar(
        title: const Text('Nova Portfolio', style:
TextStyle(fontWeight: FontWeight.bold, fontSize:
20)),
        actions: [
          IconButton(icon: Icon(isDark ?
Icons.light_mode : Icons.dark_mode), onPressed:
() => widget.onToggleTheme(!isDark)),
          Badge(
            label: const Text('2'),
            child: IconButton(icon: const
Icon(Icons.notifications_outlined), onPressed: ()
=> _toast('2 Notifications')),
          ),
          const SizedBox(width: 8),
          const Padding(
            padding: EdgeInsets.only(right: 12),
            child: CircleAvatar(radius: 16,
backgroundColor: Color(0xFF6366F1), child:
Icon(Icons.person, size: 18, color: Colors.white)),
          ),
        ],
      ),

      drawer: Drawer(
        child: ListView(
          padding: EdgeInsets.zero,
          children: [
            UserAccountsDrawerHeader(
              accountName: const Text('Parth
Rodrigues', style: TextStyle(fontWeight:
FontWeight.bold)),
              accountEmail: const
Text('parth@novawallet.io'),
              currentAccountPicture: const
CircleAvatar(backgroundColor: Colors.white,
child: Icon(Icons.account_balance_wallet, color:
Color(0xFF6366F1), size: 32)),
              decoration: const BoxDecoration(gradient:
LinearGradient(colors: [Color(0xFF4F46E5),
Color(0xFF7C3AED)])),
            ),
            ListTile(leading: const
Icon(Icons.space_dashboard_rounded), title: const
Text('Overview'), onTap: () =>
Navigator.pop(context)),
            ListTile(leading: const
Icon(Icons.show_chart_rounded), title: const
Text('Markets'), onTap: () =>
Navigator.pop(context)),
            ListTile(leading: const
Icon(Icons.swap_horiz_rounded), title: const
Text('Transactions'), onTap: () =>
Navigator.pop(context)),
            const Divider(),
            SwitchListTile(
              secondary: const
Icon(Icons.dark_mode_rounded),
              title: const Text('Dark Mode'),
              value: widget.isDark,
              onChanged: (v) { Navigator.pop(context);
widget.onToggleTheme(v); },
            ),
          ],
        ),
      ),

      body: SingleChildScrollView(
        padding: const EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment:
CrossAxisAlignment.start,
          children: [
            // Search Input
            TextField(
              decoration: InputDecoration(
                hintText: 'Search assets, tokens,
markets...',
                prefixIcon: const
Icon(Icons.search_rounded),
```

```
            filled: true,
            fillColor: isDark ? const
Color(0xFF1E293B) : Colors.white,
            border:
OutlineInputBorder(borderRadius:
BorderRadius.circular(16), borderSide:
BorderSide.none),
          ),
        ),
        const SizedBox(height: 16),

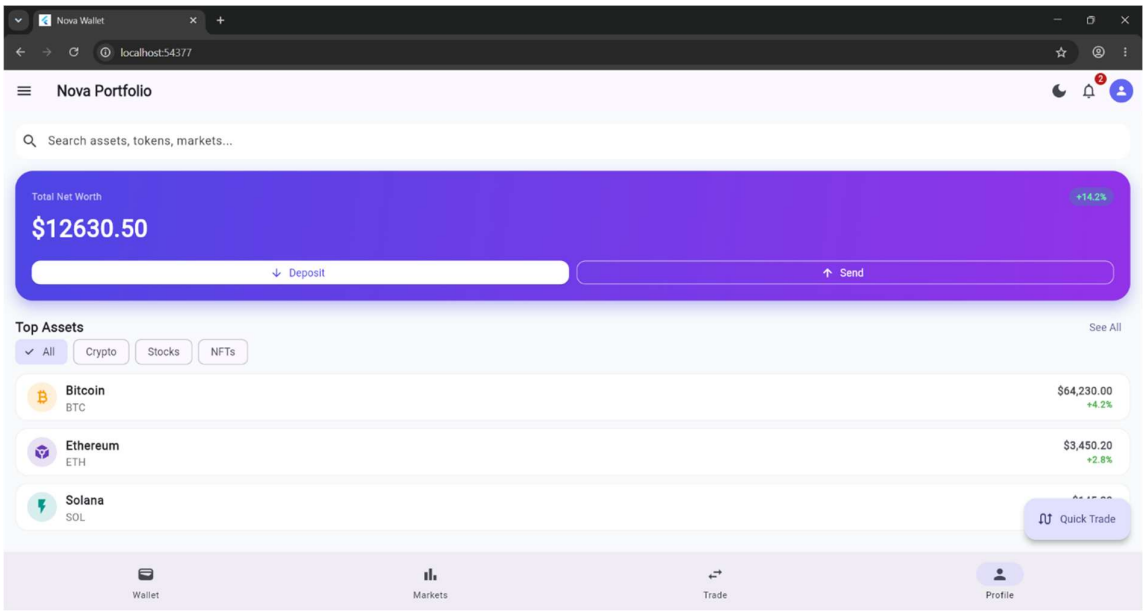
        // Hero Balance Card
        Container(
          width: double.infinity,
          padding: const EdgeInsets.all(22),
          decoration: BoxDecoration(
            gradient: const LinearGradient(colors:
[Color(0xFF4F46E5), Color(0xFF9333EA)],
begin: Alignment.topLeft, end:
Alignment.bottomRight),
            borderRadius: BorderRadius.circular(24),
            boxShadow: [BoxShadow(color: const
Color(0xFF4F46E5).withValues(alpha: 0.35),
blurRadius: 16, offset: const Offset(0, 8))],
          ),
          child: Column(
            crossAxisAlignment:
CrossAxisAlignment.start,
            children: [
              Row(
                mainAxisAlignment:
MainAxisAlignment.spaceBetween,
                children: [
                  const Text('Total Net Worth', style:
TextStyle(color: Colors.white70, fontSize: 13,
fontWeight: FontWeight.w500)),
                  Container(
                    padding: const
EdgeInsets.symmetric(horizontal: 10, vertical: 4),
                    decoration: BoxDecoration(color:
const Color(0xFF10B981).withValues(alpha:
0.25), borderRadius: BorderRadius.circular(20)),
                    child: const Text('+14.2%', style:
TextStyle(color: Colors.greenAccent, fontSize: 12,
fontWeight: FontWeight.bold)),
                  ),
                ],
              ),
              const SizedBox(height: 8),
              Text('\$${_balance.toStringAsFixed(2)}
', style: const TextStyle(color: Colors.white,
fontSize: 32, fontWeight: FontWeight.bold,
letterSpacing: 0.5)),
              const SizedBox(height: 20),
              Row(
                children: [
                  Expanded(
                    child: ElevatedButton.icon(
                      onPressed: () { setState(() =>
_balance += 250.0); _toast('Deposit successful
(+\$250)'); },
                      icon: const
Icon(Icons.arrow_downward_rounded, size: 18),
                      label: const Text('Deposit'),
                      style:
ElevatedButton.styleFrom(backgroundColor:
Colors.white, foregroundColor: const
Color(0xFF4F46E5), shape:
RoundedRectangleBorder(borderRadius:
BorderRadius.circular(12))),
                    ),
                  ),
                  const SizedBox(width: 10),
                  Expanded(
                    child: OutlinedButton.icon(
                      onPressed: () { if (_balance >=
100) setState(() => _balance -= 100.0);
_toast('Transfer completed (-\$100)'); },
                      icon: const
Icon(Icons.arrow_upward_rounded, size: 18),
                      label: const Text('Send'),
                      style:
OutlinedButton.styleFrom(foregroundColor:
Colors.white, side: const BorderSide(color:
Colors.white70), shape:
RoundedRectangleBorder(borderRadius:
BorderRadius.circular(12))),
                    ),
                  ),
                ],
              ),
            ],
          ),
        ),
        const SizedBox(height: 20),

        // Category Selector Chips
        Row(
          mainAxisAlignment:
MainAxisAlignment.spaceBetween,
          children: [
            const Text('Top Assets', style:
TextStyle(fontSize: 18, fontWeight:
FontWeight.bold)),
            TextButton(onPressed: () => _toast('See
all assets'), child: const Text('See All')),
          ],
        ),
        Wrap(
          spacing: 8,
```

```
          children: _cats.map((cat) {
            final sel = _selectedCat == cat;
            return ChoiceChip(
              label: Text(cat),
              selected: sel,
              onSelected: (s) => setState(() =>
_selectedCat = cat),
              selectedColor:
theme.colorScheme.primaryContainer,
            );
          }).toList(),
        ),
        const SizedBox(height: 12),

        // Assets List
        ListView.builder(
          shrinkWrap: true,
          physics: const
NeverScrollableScrollPhysics(),
          itemCount: _assets.length,
          itemBuilder: (ctx, i) {
            final item = _assets[i];
            return Card(
              elevation: 0,
              margin: const EdgeInsets.only(bottom:
10),
              color: isDark ? const
Color(0xFF1E293B) : Colors.white,
              shape:
RoundedRectangleBorder(borderRadius:
BorderRadius.circular(16), side: BorderSide(color:
isDark ? Colors.white10 :
Colors.grey.withValues(alpha: 0.15))),
              child: ListTile(
                leading:
CircleAvatar(backgroundColor: (item['color'] as
Color).withValues(alpha: 0.15), child:
Icon(item['icon'] as IconData, color: item['color'] as
Color)),
                title: Text(item['name'] as String, style:
const TextStyle(fontWeight: FontWeight.bold)),
                subtitle: Text(item['symbol'] as String,
style: TextStyle(color: isDark ? Colors.grey[400] :
Colors.grey[600])),
                trailing: Column(
                  mainAxisAlignment:
MainAxisAlignment.center,
                  crossAxisAlignment:
CrossAxisAlignment.end,
                  children: [
                    Text(item['val'] as String, style:
const TextStyle(fontWeight: FontWeight.bold,
fontSize: 14)),
                    Text(item['change'] as String, style:
TextStyle(color: item['up'] as bool ? Colors.green :
Colors.red, fontSize: 12, fontWeight:
FontWeight.w600)),
                  ],
                ),
              ),
            );
          },
        ),
      ],
    ),
  ),

  floatingActionButton:
FloatingActionButton.extended(
    onPressed: () => _toast('Quick Trade
Executed'),
    icon: const Icon(Icons.swap_calls_rounded),
    label: const Text('Quick Trade')),
  bottomNavigationBar: NavigationBar(
    selectedIndex: _navIndex,
    onDestinationSelected: (i) => setState(() =>
_navIndex = i),
    destinations: const [
      NavigationDestination(icon:
Icon(Icons.wallet_outlined), selectedIcon:
Icon(Icons.wallet_rounded), label: 'Wallet'),
      NavigationDestination(icon:
Icon(Icons.bar_chart_outlined), selectedIcon:
Icon(Icons.bar_chart_rounded), label: 'Markets'),
      NavigationDestination(icon:
Icon(Icons.swap_horiz_outlined), selectedIcon:
Icon(Icons.swap_horiz_rounded), label: 'Trade'),
      NavigationDestination(icon:
Icon(Icons.person_outline), selectedIcon:
Icon(Icons.person_rounded), label: 'Profile'),
    ],
  ),
 );
 }
}
```



Conclusion: By using Flutter's built-in Material 3 widgets, developers can quickly construct visually stunning, reactive, and responsive cross-platform applications with minimal, maintainable code.